

Feature Inventory

everything built since the fork

The systems, content, and tooling layered on top of Berryhunter's survival foundation to turn it into **Aura** — a top-down aura-combat MMO-lite. Each feature is described from both the in-game and the technical side. The structural rename is complete: the repo is Aura end-to-end (module `github.com/RoteRiesenRobbe/aura`, binary `aurad`).

Live: a friends & family build runs at `https://aura-game.duckdns.org/` — one process serving frontend, WebSocket and skill catalog over TLS.

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FEATURES

47

PLAYER SKILLS

20 auras · 19 cooldowns · 8 passives

40

MOB TYPES

+ 10 legacy proving-grounds

10

COMBO RECIPES

9

FACTIONS

22

EFFECT TYPES

2

ZONES

815 props · 399 spawns

1

WORLD BOSS

Generated 2026-07-22 · state = committed `main` (`b6c1d018`) · Legend: **GAME** in-game perspective **TECH** implementation · **NEW** added since the 2026-07-19 edition

1 Skill component framework — the spine of everything

GAME Every player and mob carries a spellbook and a loadout: three equipped auras but **exactly one active at a time** (a field around you), three always-on passives, three cooldown abilities. Switch auras mid-fight, spend/refund points, level skills.

TECH `skills/component.go` — `SkillComponent` (3/3/3 slots, active slot, spellbook, derived stats). One resizable circular sensor per entity, not one per skill.

2 Data-driven skill definitions + effect vocabulary

GAME Each skill is composed of effects — damage, heal, slow, resist, shield, damage-over-time, heal-over-time, summon, taunt, detaunt, light, recall, dash, revive, haste, stat buff — with per-level scaling on damage / heal / radius / target-count / tick-rate.

TECH `skills/definition.go` — 22 `EffectType`s parsed from `api/skills/*.json` with strict per-type key allowlists (hard-fail on typos). Three categories: `active_aura` / `passive` / `cooldown`.

3 Skill application system

GAME Makes auras pulse on their tick cadence, fires cooldowns (player input, or auto for mobs) with a cast-time wind-up bar, rolls crits/variance, applies no-friendly-fire and target caps.

TECH `sys/skills.go` (the big one). Per-tick sensor resize, per-effect intervals, player↔mob damage paths, cast wind-up/interrupt, `mayHarm` hostility gate, faction-relative target eligibility.

4 Capped, no-aim targeting

GAME No manual aiming — each aura picks its own targets by rule (nearest / lowest-% health / all), capped to N. Positioning decides who gets hit.

TECH `sys/targeting.go` — range filter → selector sort → take N. `selector` / `maxTargets` (+per-level) on damage/heal/instant effects.

5 Transient buff / debuff store

GAME Burns keep ticking after you leave the aura, shields absorb hits, slows / resists / haste linger.

TECH `skills/buffs.go` — one generic `Bufs` container per entity, keyed by source skill; `AbsorbShield`, `ResistMultiplier`, `SlowFraction`, `TickRateFactor`.

6 HP, resistances & damage tags, stat variance

GAME Absolute HP pools; mobs resist / are immune to specific damage types (string tags, incl. `"*"` wildcard); harvest & gate obstacles only take damage from the "right" aura; per-hit and spawn-HP variance.

TECH Resist maps in `skills/resist.go`, gated damage tags, spawn-HP bands.

7 Crit as a character stat **NEW**

GAME Everyone crits: a flat base chance on every hit, raised by stacking passives (KeenEye) rather than by picking "crit skills". Damage-over-time gets a flat compensation instead of rolling crits.

TECH `StatCritChance` as a derived stat applied in `rollHitDamage`; base chance from config, `critFactor` multiplier, authored crit pairs removed (crit rework v2).

Progression & unlocks

8 Spellbook, skill points, free respec

GAME Discover skills (never lost on death), spend points earned per level to raise levels, freely respec down to discovery level.

TECH `skills/component.go` — spellbook, point accounting, raise/lower level; skills survive death via carried component in `sys/state.go`.

9 Five unlock sources

GAME Skills come from **milestone levels**, **mob kill-drops** (chance, shared with recent healers), **NPC teaching**, **recipes**, and (scoped) character sacrifice.

TECH `skills/milestones.go` fed by `api/milestones/`; `MobUnlock` in `items/mobs/definitions.go`; `sys/npc.go`; `skills/recipe*.go`.

10 Aura combinations / recipes

GAME Secret, undocumented recipes — hold a set of skills at required levels and a capstone auto-unlocks (Vanguard trio → Warbanner). Community-discovered.

TECH `skills/recipe.go` + `recipe_apply.go` — cascading, idempotent threshold check on every discovery / level-up.

11 XP for all participants + a derived XP curve

GAME Every damager (and their recent healers) gets full XP on a kill — no groups, no split, no grief. Each mob's XP value is derived from how many of it you can actually kill per hour, so any band levels at a comparable pace.

TECH `model/mob/mob.go` `rewardPlayer` — participant + recent-healer tracking, then the recipe cascade. XP values authored from the band rule $3600 \times 1.15^{(cL-1)} \div \text{measured kills/h}$.

Mobs & AI

12-17

12 Mob AI depth

GAME Threat table (per-player aggro, taunt / detaunt), state-dependent leash + walk-home, out-of-combat regen, flee-when-low, auras-off-until-aggroed. Idle archetypes: stationary / local wander / route patrol with evade-return, plus obstacle steering that commits to a detour instead of jittering.

TECH `model/mob/mob.go`, `patrol.go`, `steering.go`, `stuck.go` — threat ops, invuln/flee overrides, TTL, buff applicers, chase stuck-watchdog.

13 Faction system

GAME Players + summons are "aligned"; content can declare factions friendly-to-players (never harmable, still fight other factions), plus mob-vs-mob hostility.

TECH `model/faction.go` — `HostilityGate.MayHarm`, `PlayerFriendly`; enforced in `sys/skills.go`; rosters in `api/factions/`.

14 Summons / companions

GAME Cooldowns spawn owned, player-aligned mobs: stationary totems or following companions that defend-then-assist and expire on a timer; owner-attributed XP.

TECH `sys/skills.go` `spawnSummon`; `model/mob/companion.go` — follow ring, tether, owner combat-signal acquisition, teleport catch-up.

15 Support mob (seek-healer)

GAME Friendly healer mob seeks the most-wounded ally, chases it; heal aura toggles on only while healing.

TECH `model/mob/healer.go` — reuses the aggro machinery with allies instead of enemies.

16 Encounter controller + Orc Warlord world boss

GAME First designed boss: invulnerable while either Warbanner totem stands; grunt waves at 66 / 33%; banners replant; wiping resets it; killing it broadcasts the credited players server-wide, empties the arena, and respawns him ~5 min later.

TECH `encounter/system.go` (ECS lifecycle hooks, spawn/despawn/announce); `encounter/warlord.go` (script); positions from zone anchors.

17 Authored mob power model — curve level + tier NEW

GAME Every combat mob is authored as "a level-N normal / elite / boss" instead of hand-typed HP: tiers mean something concrete (elite $\leq$ 25% facetankable, boss kills the standing bot), and each zone band has a shape the roster must fit.

TECH Raw `maxHealth` hard-fails; stats derive from curve level $\times$ tier baseline. Deterministic band asserts in `cmd/simharness/guardrail_test.go` (Z1 / Z2 / farm bands, front exempt).

World & atmosphere

18-22

18 Authored zone system

GAME The procedural world is replaced by hand-authored zones: bounds, terrain, props, spawns, campfires, dark areas, teaching NPCs, and named encounter anchors; `-zone` selectable. The shipped world is 815 props, 399 spawns, 35 dark areas, 14 NPCs, 5 campfires.

TECH `world/zone.go` — `LoadZoneFS` with strict validation; anchors, campfires, dark areas, spawn archetypes, prop registry.

19 NPC teaching / lore

GAME Peaceful, unattackable NPCs teach ordered skills on approach (level-gated, "too low" line) or speak lore / sign-post lines — including the village arrival beat, where a Town Crier hands out the first combat aura and Recall.

TECH `sys/npc.go` — rising-edge proximity; `onApproach` grants + recipe cascade; `speak` reuses the chat wire.

20 Campfire recovery, respawn, revive & recall

GAME Campfires heal a % of max HP and, after a short dwell, become your respawn point; fresh players spawn at a starting campfire; Recall teleports you back; Revive raises a downed player at their corpse.

TECH `sys/state.go` — campfire anchors, dwell tracking, default spawn, `tryRespawn`, `ReviveAtCorpse`, `AnchorOf`.

21 Light, darkness & a readable night

GAME Authored dark-area circles independent of day/night; a light aura or Torch passive punches a lit hole — a real slot trade-off (light vs. combat aura). Night dims the world without swallowing characters: nameplates stay legible above the tint.

TECH `light_aura` effect + `light_radius` wire (max over active aura and passives); client `DarknessOverlay.ts`; `DayCycle.ts` night-filter layer list is derived (new layers are night-correct by default) with an unfiltered `namePlates` overlay.

22 Chat / server broadcast

GAME Proximity speech bubbles plus server-wide system banners (boss beats, `ANNOUNCE`).

TECH `sys/chat/system.go` — `Broadcast` (sender id 0 = system banner) + proximity say via viewport.

Client / UI

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23 Spellbook UI

GAME Scrollable Auras / Passives / Cooldowns panel, level badges with +/- spend buttons gated by skill points, gold glow on genuine unlocks (level-ups & baseline never false-trigger).

TECH `HUD.ts` setup/update + SimpleBar scroll; diff logic against known spellbook ids/levels.

24 Ability bar & loadout

GAME Aura slots (hotkeys 1-3, one active / toggle-to-Nothing), passive slots, cooldown slots (Q/E/F) with remaining-seconds readout and a cast bar stacked above them; loadout editing locked in combat with feedback.

TECH `HUD.ts`, `Controls.ts`, `InputMessage.ts` — server-authoritative with optimistic-highlight reconciliation.

25 Category aura rings — what is this field doing to me? **NEW**

GAME The translucent field around a caster is colour-coded by what it does — damage, burn, heal, shield, slow, light, resist — with one band per category stacked inward from the aura's edge, so the outer circle always means "this is the range". An edge glow pulses in time with each tick and speeds up under haste; per-tick hit VFX plus floating damage / heal / XP numbers.

TECH Server-derived `skills.AuraCategory` bitmask on the wire (`aura_category`, Character + Mob); one shared `AuraRingStack` in `AuraRings.ts` replaced three ring implementations; `AuraTickIndicator.ts` for the tick phase.

26 Applied-effect pips & mob tier frames **NEW**

GAME Small coloured pips under the overhead bar show what is currently applied *to* an entity — burning, slowed, regenerating, resisting, hasted — in the same colour language as the rings. Elites wear a silver frame, bosses a gold one, so "should I fight this" is readable before the health bar moves.

TECH `applied_effects` ubyte on both tables (the received-status mirror of `aura_category`), compile-enforced from the buff payloads; shared `EffectPips.ts` + `AURA_CATEGORY_COLORS`; tier frames in `Mobs.ts`.

27 Skill catalog endpoint + hover tooltips **NEW**

GAME Hovering any skill — spellbook, ability bar — shows a full tooltip: what it does, the numbers at your current level and what the next level buys, radius, targets, cadence.

TECH `GET /skills` serves the **parsed** registry as JSON (`skills/catalog.go`), fetched once at startup. This deleted the hand-synced name/category/maxLevel maps in `Skills.ts` — the client can no longer drift from server balance. Tooltips in `SkillTooltip.ts`.

28 AlertBanner

GAME Upper-third single-slot queued banner for server announcements + gold new-skill unlock notices.

TECH `alert-banner/AlertBanner.ts` — queue + `show(text, kind)`; fed from `EntityMessage` id 0.

29 Reload-safe session — reconnect token **NEW**

GAME Refreshing the tab (or a brief disconnect) no longer costs you your character: you come back as the same player, same level, same spellbook — for roughly ten minutes.

TECH Append-only `Join.reconnect_token` / `Accept.reconnect_token`; server-side `stashByToken` in `sys/state.go` (disconnect stashes the character and holds the name, TTL ~10 min); client `Session.ts` in `sessionStorage` with auto-rejoin.

30 Portrait rotation freeze

GAME Avatars are fixed-facing portraits that never spin to the pointer (local player included).

TECH `Character.ts` — `setRotation` forces downward facing; held items still transform.

31 Zone editor dev panel

GAME In-game placement / editing of props, spawns, campfires, dark areas, NPCs, and anchors, with hit-testing, a registry-validated teaching-skill dropdown, and mob entries that show curve level + tier while you place them.

TECH `zone-editor/ZoneEditor.ts` + `ZoneModel.ts` — per-kind place/update/remove/hitTest, marker drawing, export back to zone JSON.

32 Mob roster art

GAME ~50 new EntityTypes / SVGs: wildlife and dire variants, kobolds (melee / ranged), spiders + venom, bandits (ranged / healer / elite / pyromancer), army & orc faction incl. the Warlord boss, fire elementals, player summons, village NPCs, and hazards (PoisonPool, Rockfall, Bramble, SpikeBarricade, Brazier, Campfire, Turnip).

TECH `Mobs.ts` (~47 classes, each registers an SVG); NPCs ride the Resource path in `Resources.ts`.

Tooling & operations

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33 Dev cheats

GAME Token-gated testing commands: `GOD`, `WARP x y`, `SPEED [factor|off]`, `XP <n>`, `SKILL <name>`, `KILL`, `DAMAGE <pct>`, `THREAT [id]`, `ANNOUNCE <text>`, `PING`.

TECH `sys/cmd/cmd.go` `CommandSystem` — static map + space/announcer-bound closures. On the live server the token list lives outside the deploy bundle, so players never get one.

34 Content pipeline — edit JSON, restart, play **NEW**

GAME Every tunable — skills, mobs, factions, recipes, milestones, props, zones — is content, not code. Balance passes are a JSON edit and a restart, on the dev box or on the live server.

TECH `aurad -content ../api` loads from the repo tree instead of the embedded copy; the milestone table moved into `api/milestones/` to close the last gap, so content deploys skip the rebuild entirely (`deploy.sh --content-only`).

35 Simulation harness

GAME Headless balancing explorer that drives the real ECS: TTK / TTD, kills-per-level, 1-vs-N matrix, stand-still-bot facetank thresholds, plus a `-serve` web UI.

TECH `cmd/simharness` — full-roster presets incl. dot auras, plus `guardrail_test.go`: deterministic asserts that the shipped roster stays inside its band and that the power ceiling keeps its ordering.

36 Load-test harness **NEW**

GAME Answers "how many players can this box hold": a bot swarm joins over the real protocol at the real tick rate and ramps population until the loop falls behind.

TECH `cmd/loadbot` (ramp steps, snapshots/sec + kB/s per bot) against `-profile's /tickstats` ring (p50/p95/p99 vs the 33 ms budget). Profiler is opt-in and never enabled in production.

37 One-command deploy + live playtest server **NEW**

GAME The game is playable on the open internet by anyone with the link — testers just open the URL, no install, no account.

TECH A single `aurad` process serves frontend, `/game` WebSocket and `/skills` with built-in Let's Encrypt TLS + port-80 redirect, under systemd with restart-on-failure; `devops/deploy.sh` (full / `--content-only`) + runbook.

Worth flagging as NOT built, despite appearances

Aura line-of-sight was deliberately cut — auras pass through walls; walls only block movement (the client's light/darkness hole is a soft presentational model, not raycast occlusion).

There is no persistence. Characters live in server memory only: the reconnect token survives a reload for ~10 minutes, but any server restart wipes every character — which is why live deploys are announced in-game first. Accounts & persistence is the next system, with the character-sacrifice loop as its first consumer.

Roster snapshot (committed `main`, `b6c1d018`): 47 player skills (20 auras / 19 cooldowns / 8 passives) · 35 mob skills · 40 world mobs + 10 legacy proving-grounds · 9 factions · 10 combination recipes · 2 zones + 1 boss encounter · 22 effect types. All numbers are placeholders until marked final — and "final" here means first-pass settled, not frozen.